Module 09: Active Directory Penetration Testing

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# Exercise 07: Enumerating Unconstrained Delegation
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Scenario

Kerberos **Unconstrained Delegation** allows a designated service or computer account to impersonate any domain user across any service in an Active Directory forest. When a user authenticates to a server configured with unconstrained delegation (UserAccountControl bit `TRUSTED_FOR_DELEGATION` / `0x80000`), the domain controller includes a copy of the user's unencrypted **Ticket Granting Ticket (TGT)** inside the Service Ticket (TGS). The compromise of a computer with unconstrained delegation enables attackers to harvest cached domain administrator TGTs from memory and compromise the entire domain.

In this exercise, we access **2019DC**, inspect Active Directory Users and Computers (ADUC) delegation settings for `SERVER2019`, `WIN2016`, and `WINDOWS11`, configure Unconstrained Delegation on `WINDOWS11` ("Trust this computer for delegation to any service (Kerberos only)"), import **PowerView** (`PowerView.ps1`), execute **Get-NetComputer -Unconstrained** to identify unconstrained computer accounts (e.g. `windows11$`), enumerate Group Policy Objects in SYSVOL (`\\server2019\SYSVOL\LPT.com\Policies\`), and inspect SYSVOL access control lists using **Get-Acl**.

Objective

The objective of this lab is to help students learn how to:
- Inspect Kerberos delegation settings in ADUC across Domain Controllers and member workstations
- Enable Kerberos Unconstrained Delegation on `WINDOWS11` target machine
- Circumvent PowerShell script execution restrictions using **set-executionpolicy Bypass**
- Load PowerView cmdlet library via **Import-Module .\Powerview.ps1**
- Enumerate all domain computers configured with unconstrained delegation using **Get-NetComputer -Unconstrained**
- Retrieve the `samAccountName` property of member workstations configured with unconstrained delegation (e.g., **windows11$**)
- Enumerate SYSVOL Group Policy Object paths via **Get-ChildItem -Path \server2019\SYSVOL\LPT.com\Policies**
- Audit SYSVOL policy directory access control permissions using **Get-Acl -Path \server2019\SYSVOL\LPT.com\Policies\ | format-list**

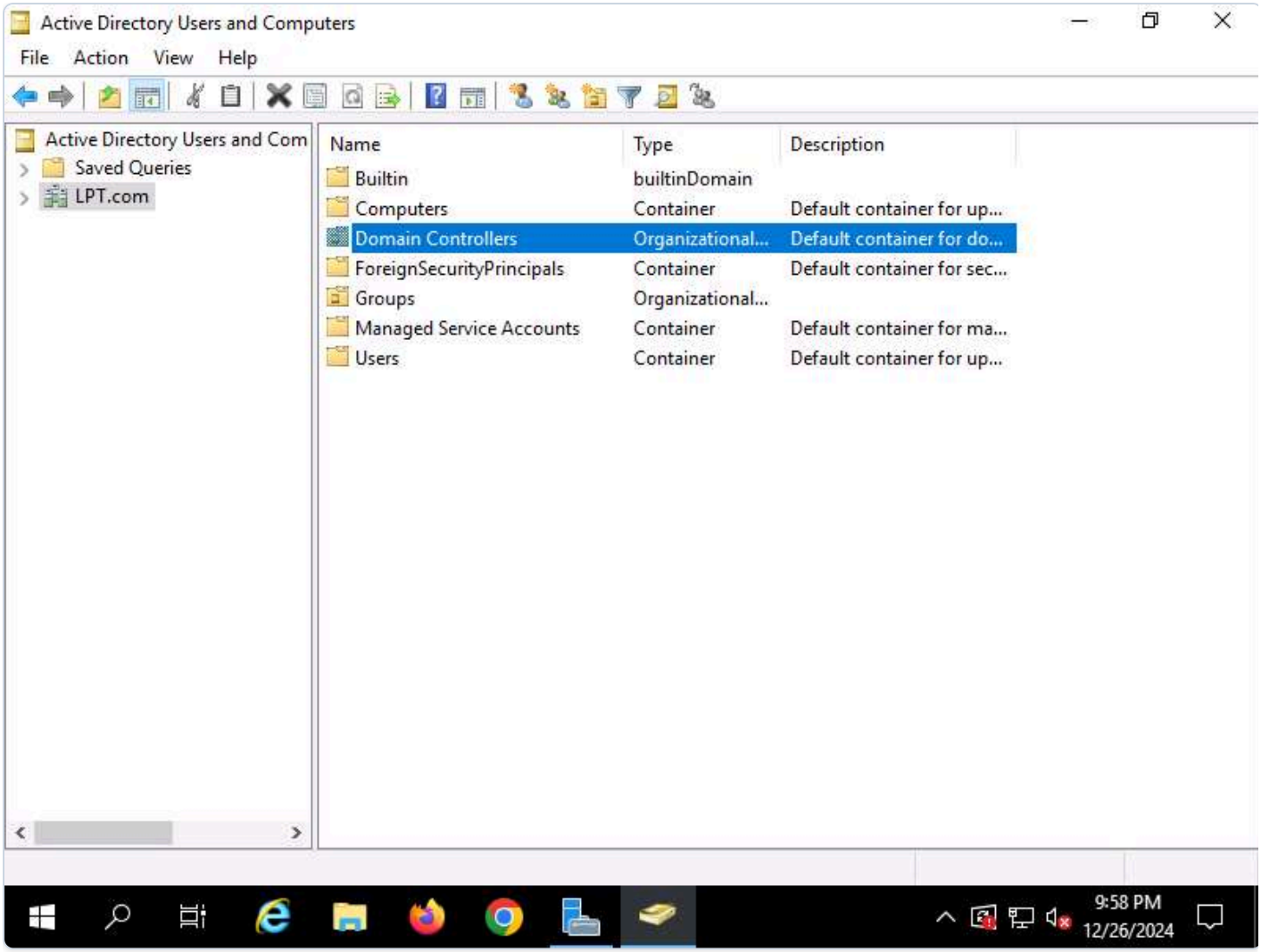
Task 1: Enable Unconstrained Delegation on WINDOWS11 and Enumerate Delegation Policies via PowerView

In this task, we log into **2019DC** (**LPT\Administrator** / **Pa$$w0rd**), inspect ADUC delegation properties, configure Unconstrained Delegation on `WINDOWS11`, run PowerView enumeration to identify `windows11$`, and inspect SYSVOL permissions.

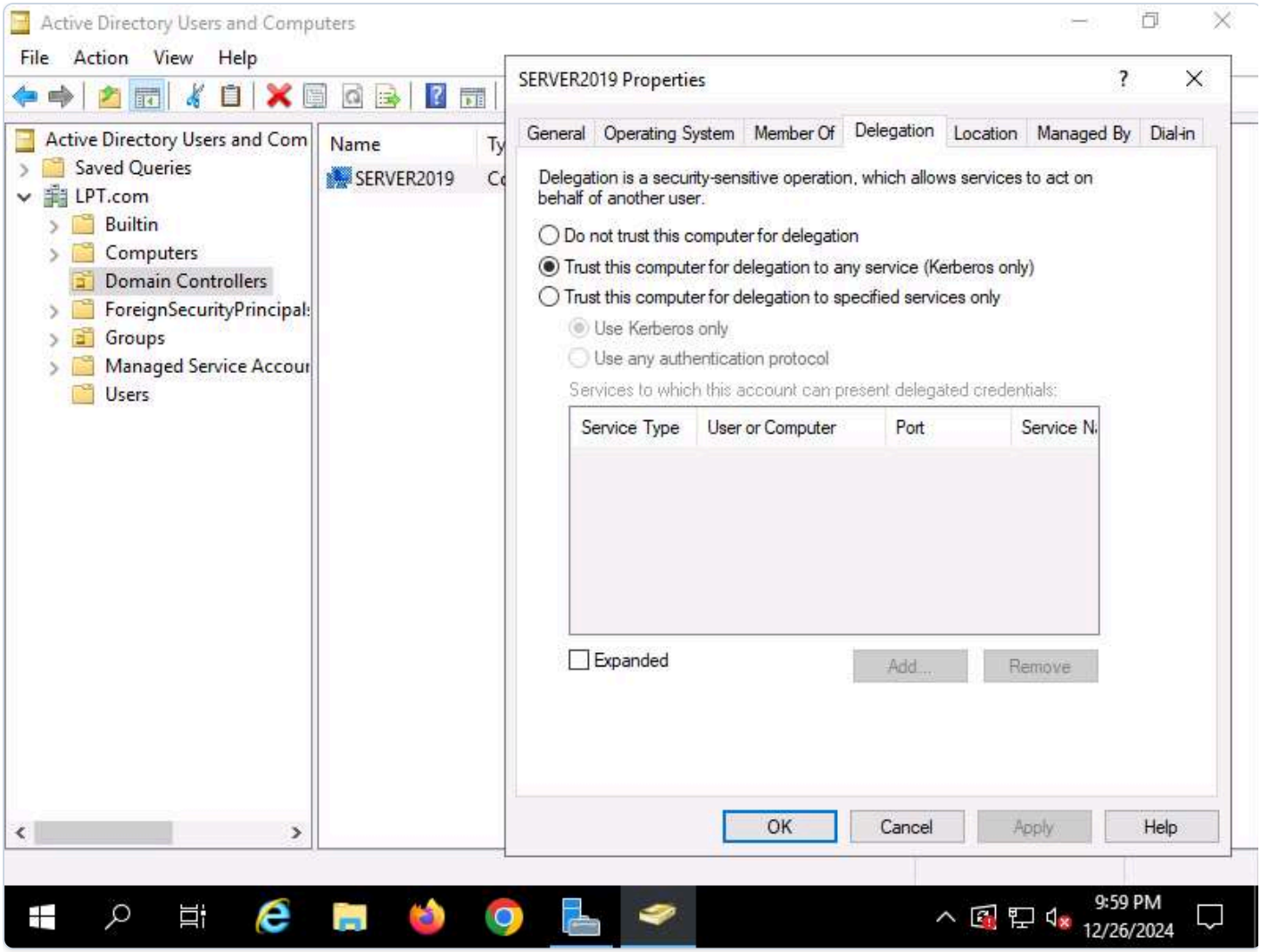
Lab Tasks

1. Switch to **2019DC** machine and log in as Domain Administrator (**LPT\Administrator** / **Pa$$w0rd**).

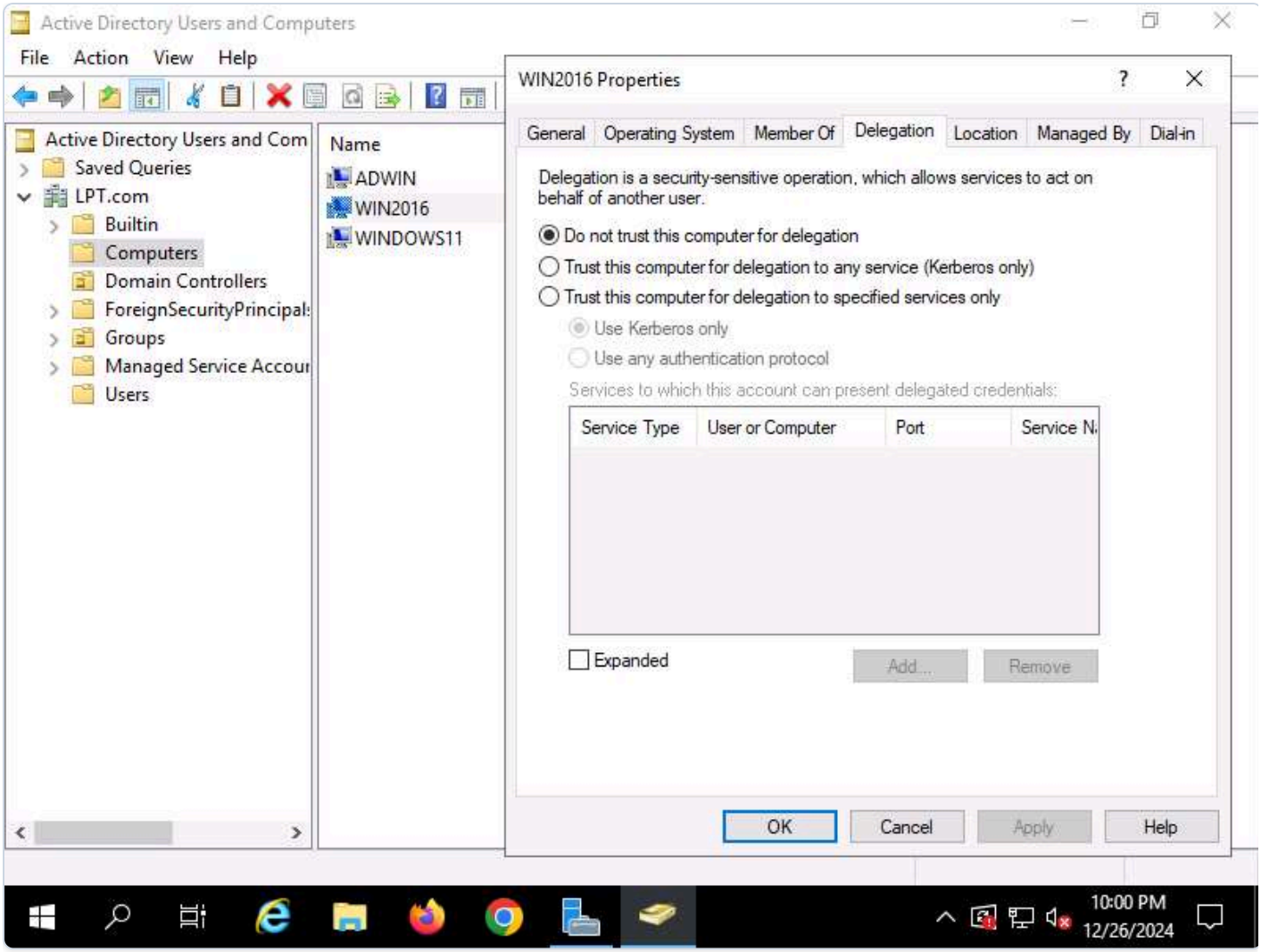
2. Open **Server Manager** $\rightarrow$ **Tools** $\rightarrow$ **Active Directory Users and Computers**. Expand `LPT.com` and open `Domain Controllers`.



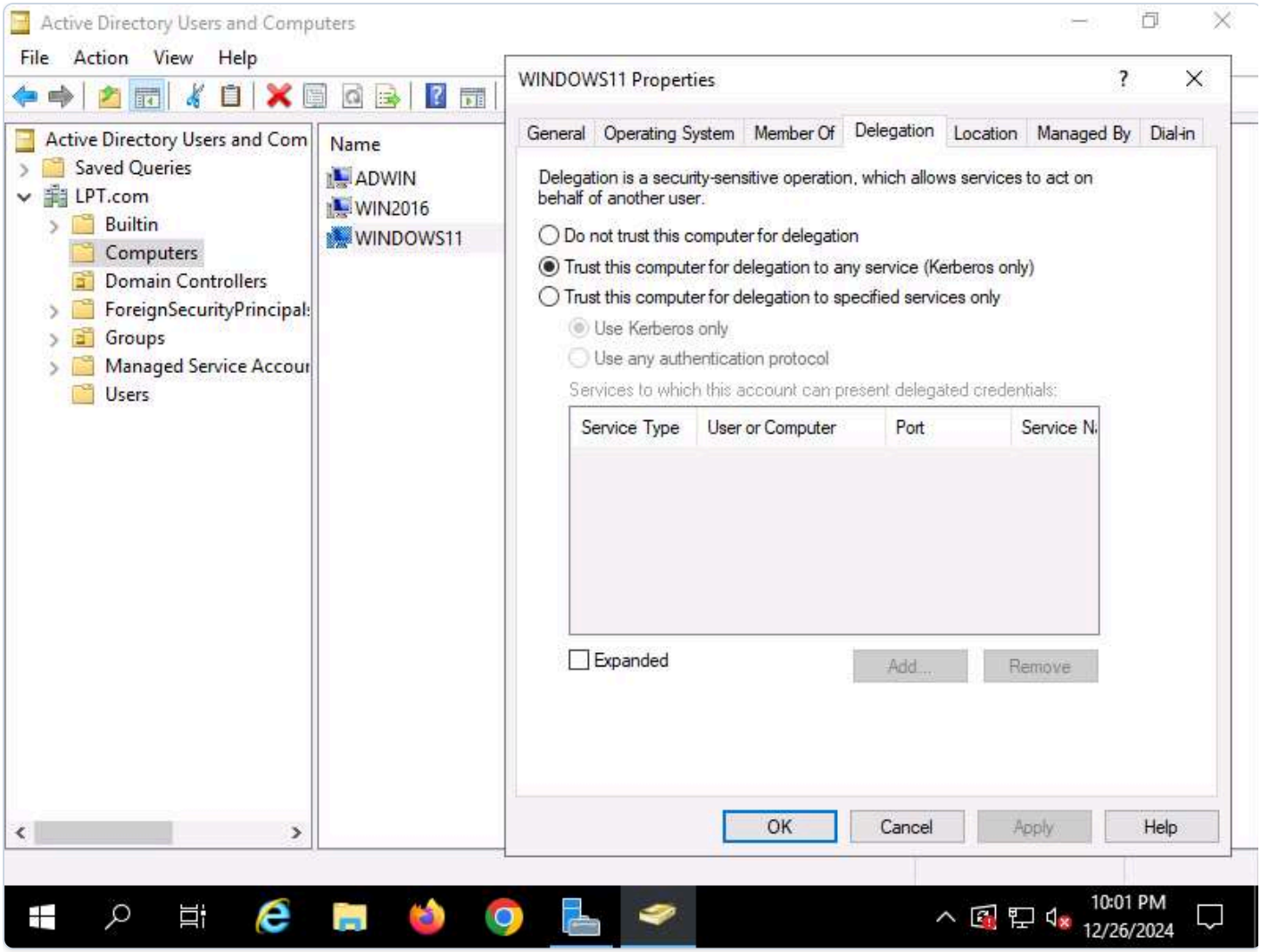
1. Right-click `SERVER2019`, select **Properties** $\rightarrow$ **Delegation** tab. Note that unconstrained delegation ("*Trust this computer for delegation to any service (Kerberos only)*") is enabled by default on Domain Controllers.



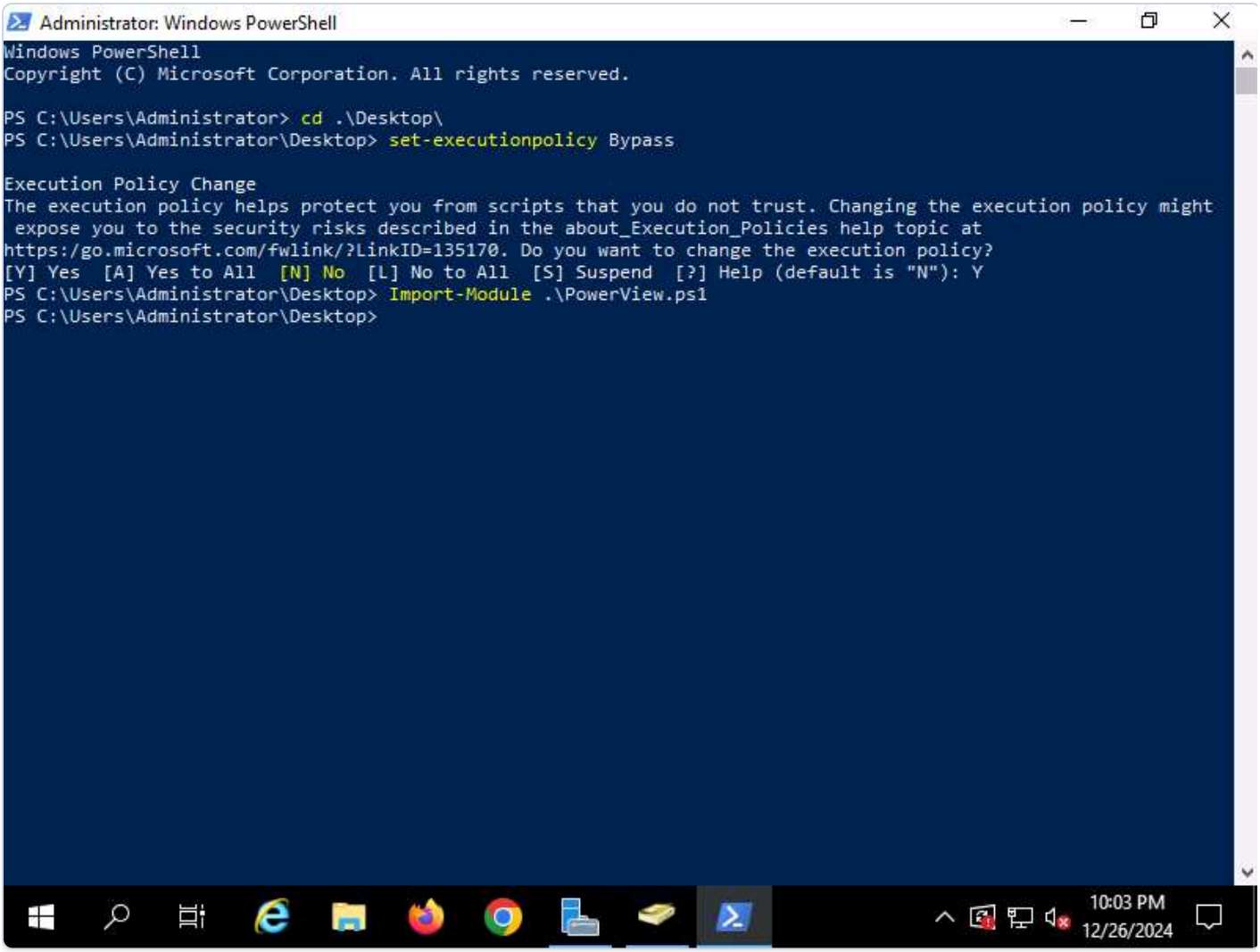
1. Navigate to `Computers`, right-click `WIN2016`, select **Properties** $\rightarrow$ **Delegation** tab. Observe that delegation is disabled by default ("*Do not trust this computer for delegation*").



1. Select `WINDOWS11` under `Computers`, open **Properties** $\rightarrow$ **Delegation** tab, select **Trust this computer for delegation to any service (Kerberos only)**, then click **Apply** and **OK**.



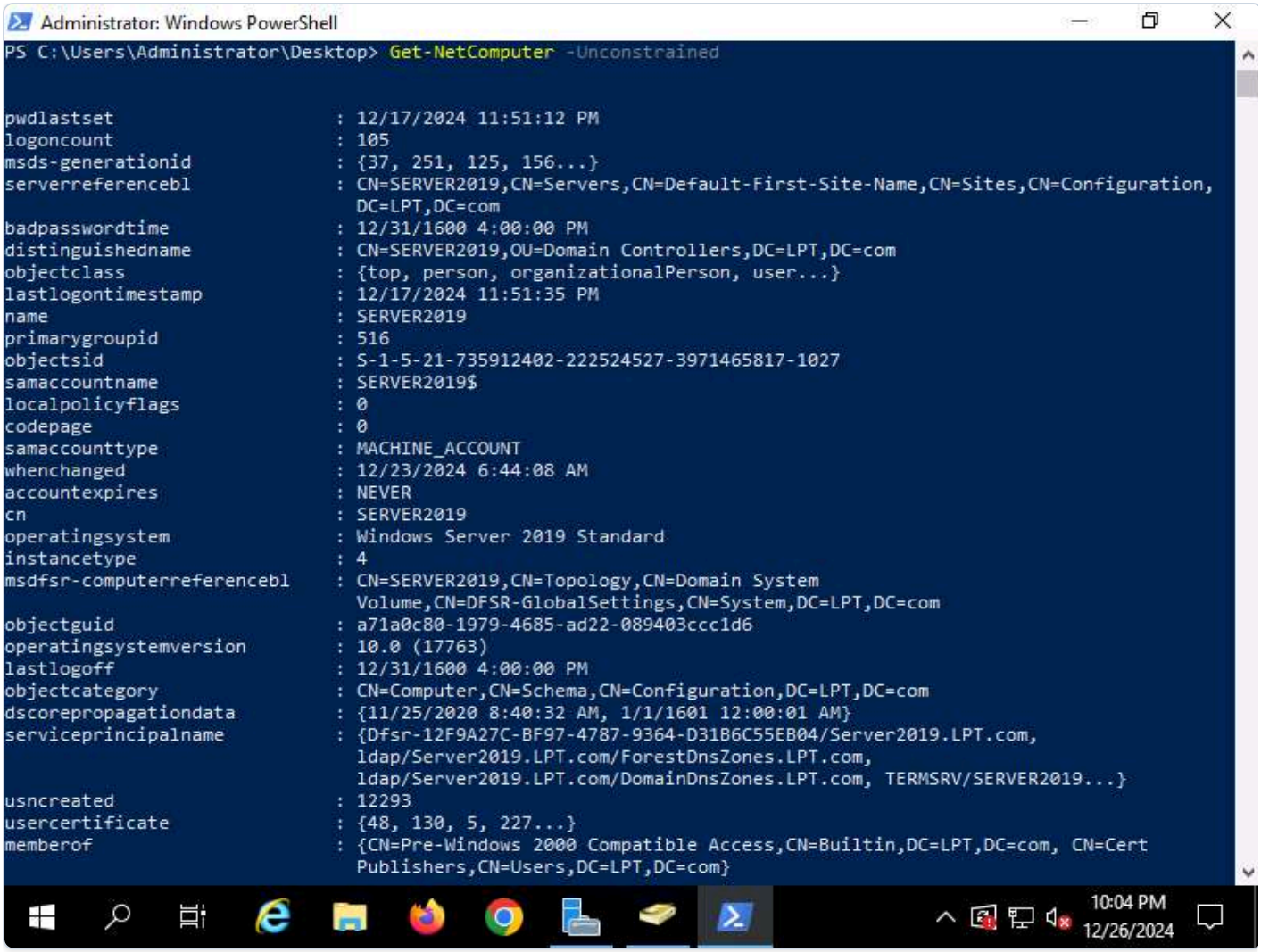
1. Open Windows PowerShell, navigate to Desktop (`cd Desktop`), bypass execution policy, and import PowerView:
   set-executionpolicy Bypass
   Import-Module .\Powerview.ps1

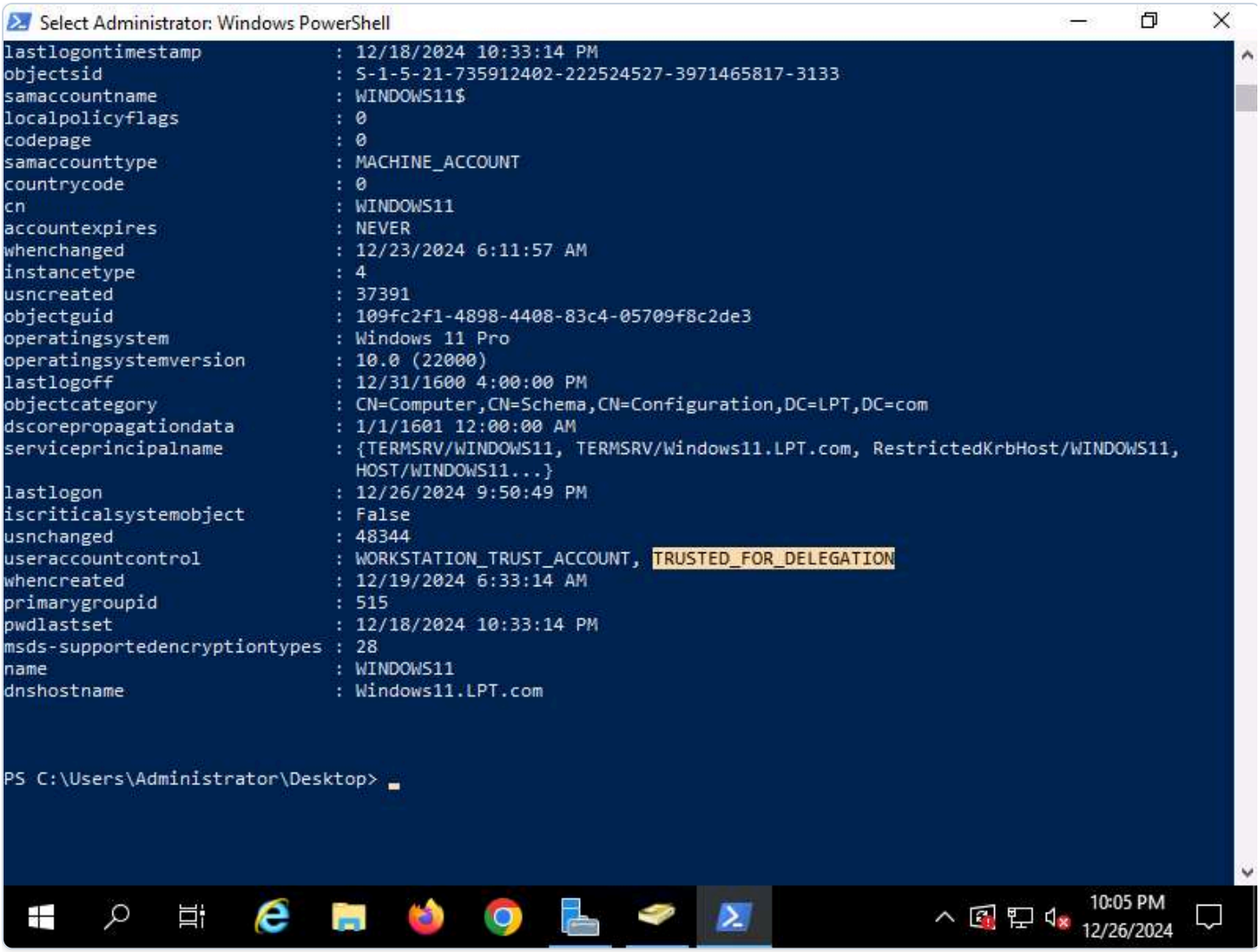


1. Enumerate all computer objects in the domain configured with Kerberos Unconstrained Delegation:

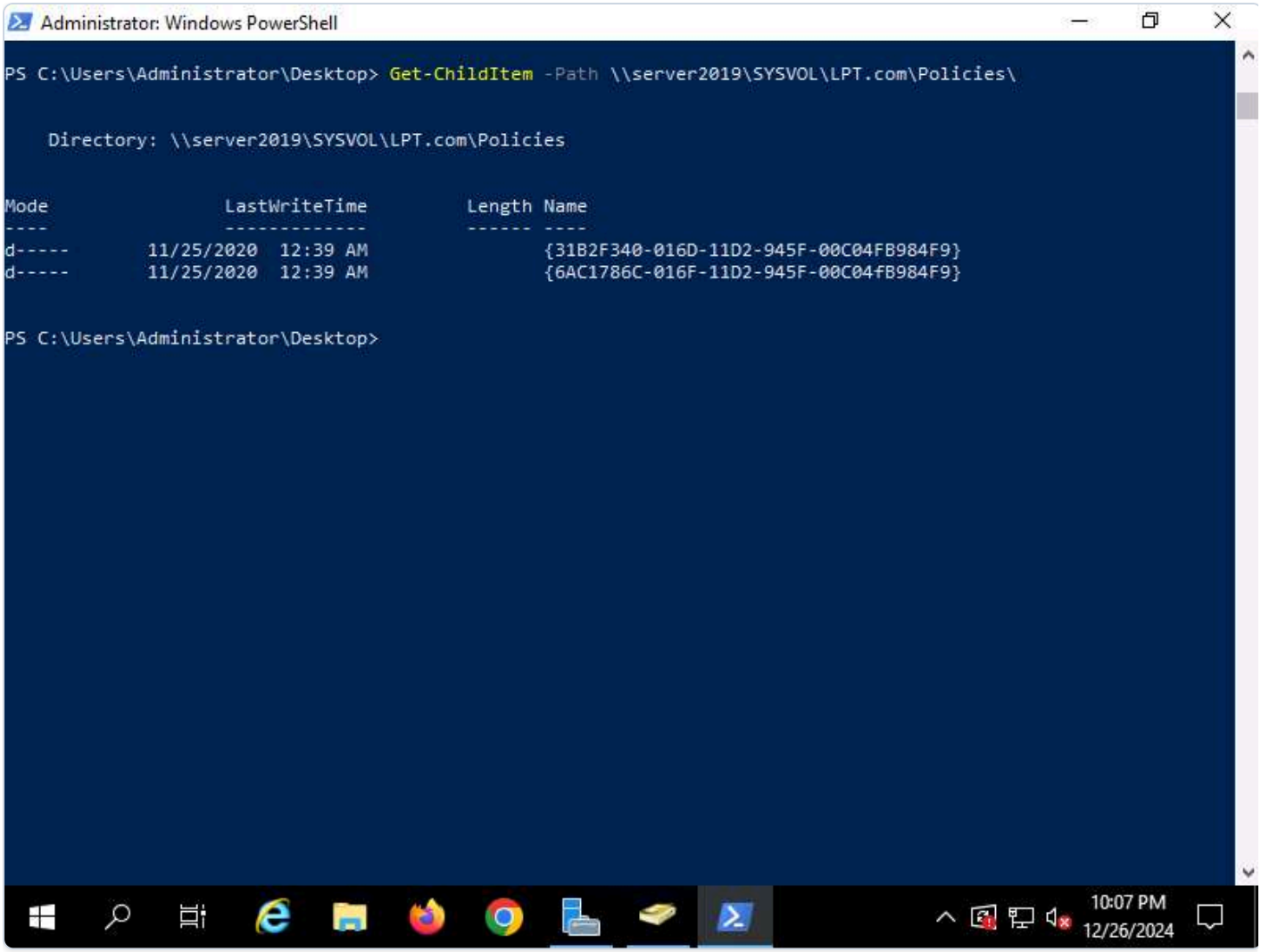
   Get-NetComputer -Unconstrained

   Observe `samAccountName` *for* `WINDOWS11`: ***windows11$***.

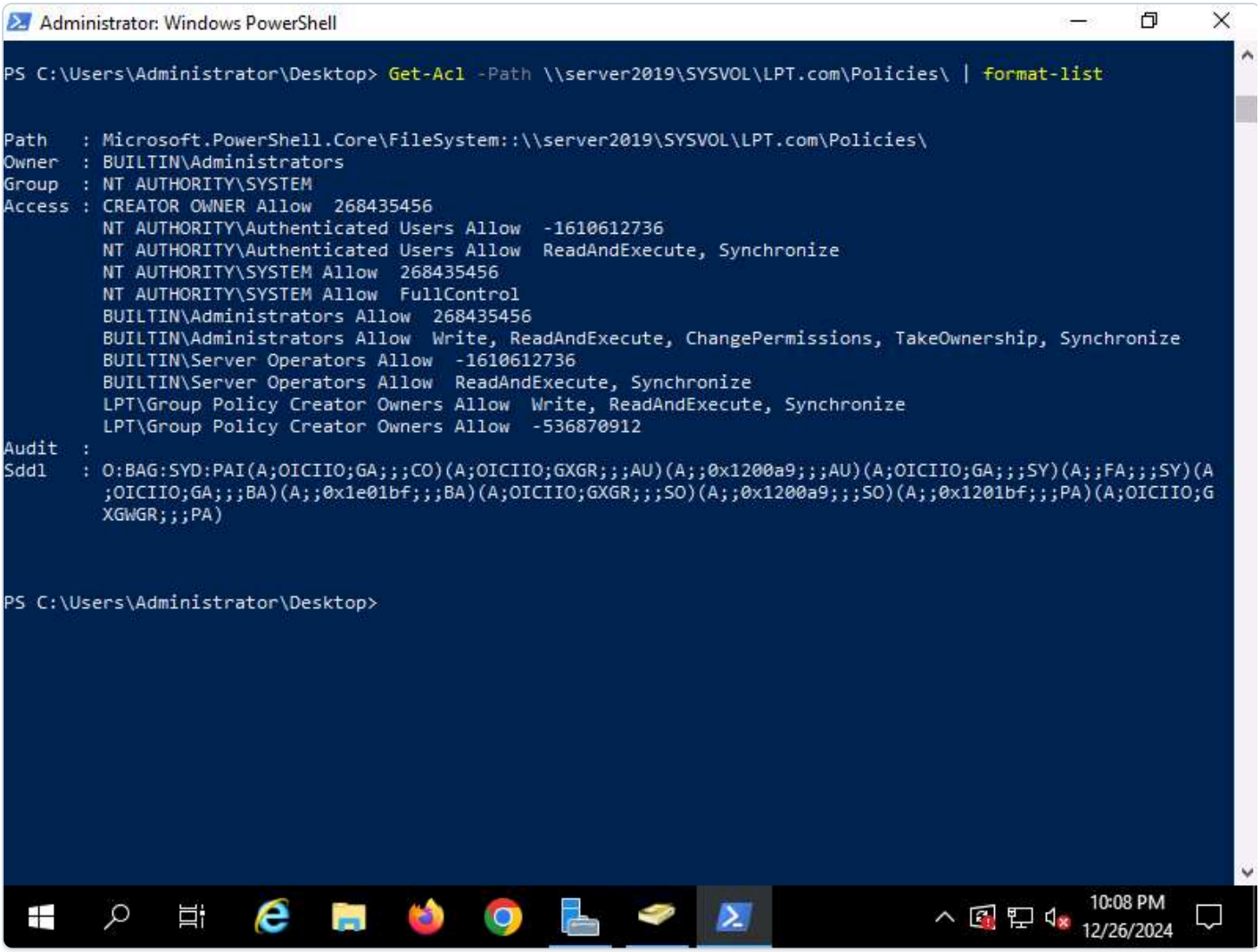

Administrator: Windows PowerShell
PS C:\Users\Administrator\Desktop> Get-NetComputer -Unconstrained
pwdlastset : 12/17/2024 11:51:12 PM
logoncount : 105
msds-generationid : {37, 251, 125, 156...}
serverreferencebl : CN=SERVER2019,CN=Servers,CN=Default-First-Site-Name,CN=Sites,CN=Configuration, DC=LPT,DC=com
badpasswordtime : 12/31/1600 4:00:00 PM
distinguishedname : CN=SERVER2019,OU=Domain Controllers,DC=LPT,DC=com
objectclass : {top, person, organizationalPerson, user...}
lastlogontimestamp : 12/17/2024 11:51:35 PM
name : SERVER2019
primarygroupid : 516
objectsid : S-1-5-21-735912402-222524527-3971465817-1027
samaccountname : SERVER2019$
localpolicyflags : 0
codepage : 0
samaccounttype : MACHINE_ACCOUNT
whenchanged : 12/23/2024 6:44:08 AM
accountexpires : NEVER
cn : SERVER2019
operatingsystem : Windows Server 2019 Standard
instancetype : 4
msdfsr-computerreferencebl : CN=SERVER2019,CN=Topology,CN=Domain System Volume,CN=DFSR-GlobalSettings,CN=System,DC=LPT,DC=com
objectguid : a71a0c80-1979-4685-ad22-089403ccc1d6
operatingsystemversion : 10.0 (17763)
lastlogoff : 12/31/1600 4:00:00 PM
objectcategory : CN=Computer,CN=Schema,CN=Configuration,DC=LPT,DC=com
dscorepropagationdata : {11/25/2020 8:40:32 AM, 1/1/1601 12:00:01 AM}
serviceprincipalname : {Dfsr-12F9A27C-BF97-4787-9364-D31B6C55EB04/Server2019.LPT.com, ldap/Server2019.LPT.com/ForestDnsZones.LPT.com, ldap/Server2019.LPT.com/DomainDnsZones.LPT.com, TERMSRV/SERVER2019...}
usncreated : 12293
usercertificate : {48, 130, 5, 227...}
memberof : {CN=Pre-Windows 2000 Compatible Access,CN=Builtin,DC=LPT,DC=com, CN=Cert Publishers,CN=Users,DC=LPT,DC=com}
10:04 PM
12/26/2024



1. Enumerate Group Policy Objects stored in the Domain Controller SYSVOL share:
   Get-ChildItem -Path \server2019\SYSVOL\LPT.com\Policies



1. Inspect Access Control Lists (ACLs) for the SYSVOL Policies folder:
   Get-Acl -Path \server2019\SYSVOL\LPT.com\Policies\ | format-list



This concludes the lab exercise.

IMPORTANT: ❓ **Question 9.7.1.1** On the 2019 Domain Controller (DC) machine, enable unconstrained delegation for the Windows 11 machine. Then, use PowerView to enumerate access control policies. Enter the samAccountName of the Windows 11 machine obtained during the enumeration process. [!SUCCESS] ✅
Answer: **windows11$**